

# Internship Progress Report

## **Theft Detection**

June – July 2025

## **Safety and Accidents Analysis**

July – August 2025

KEARNEY



# Project I Overview: Theft Detection

| Topics                                                          | Sub-Topics Covered                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Identify Non-Domestic locations registered as Domestic          | <ul style="list-style-type: none"><li>• Scraping commercial establishments.</li><li>• Proximity Analysis using BallTree.</li><li>• Flagging suspected theft cases.</li></ul>                                                                                                                                                                     |
| Identify Non-Domestic Locations Registered as Small Industries. | <p>Consumption Based:</p> <ul style="list-style-type: none"><li>• Consumption Pattern Analysis using Feature Extraction &amp; PCA.</li><li>• Identifying Anomalies between SIP and NDS locations.</li></ul> <p>Address Based:</p> <ul style="list-style-type: none"><li>• Nearest NDS to SIP, Similarity scoring, Owner info matching.</li></ul> |
| Future Prospects                                                | <ul style="list-style-type: none"><li>• Alternate approaches for fetching owner info.</li><li>• Scaling to other districts.</li><li>• Proper documentation in GitHub.</li><li>• Automation and dashboard integration.</li></ul>                                                                                                                  |

# I. Non Domestic as Domestic

## STEP 01

- ✓ Implemented a Google Maps API-based pipeline to collect commercial establishment information for a given location and radius.
- ✓ Fetched ~3500 locations in a 10 km radius from Tonk center.

- ✓ Applied the BallTree algorithm for efficient nearest-neighbor search, identifying the closest domestic and non-domestic location for each commercial establishment.
- ✓ Reduced computation time from 20–25 minutes (brute force) to just 1–2 seconds.

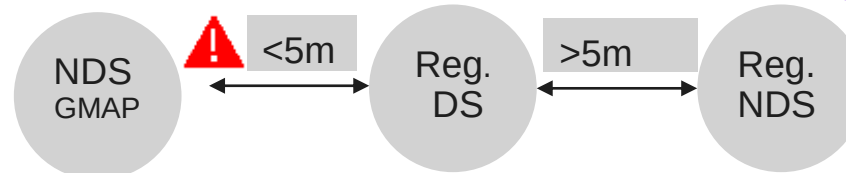
## STEP 02

## STEP 03

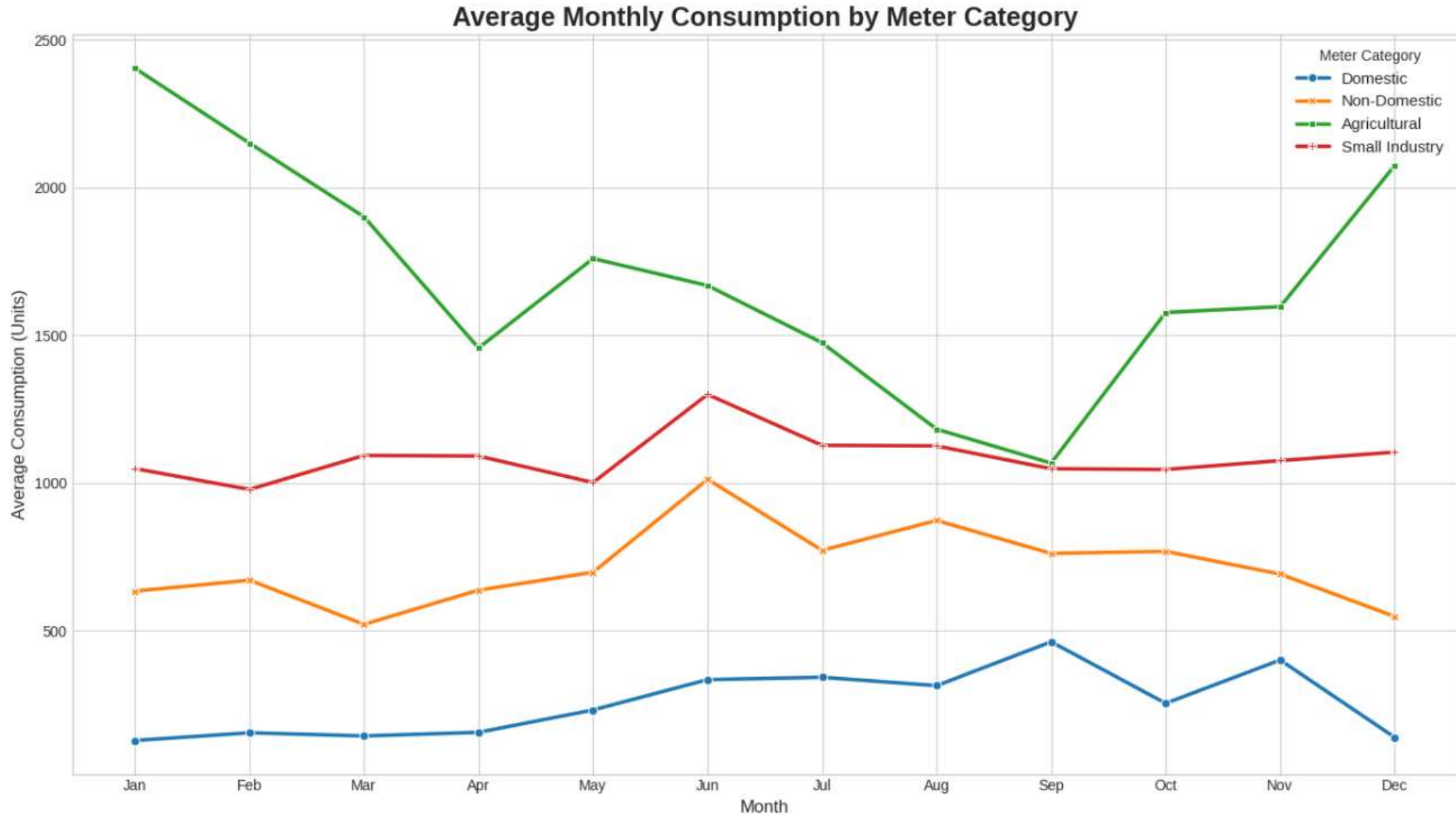
- ✓ After filtering, ~150 (4%) establishments were shortlisted as potentially high risk cases

- ✓ Flagged as potential theft when a commercial establishment is closer to a domestic site (<5m) than to any non-domestic site (>5m)

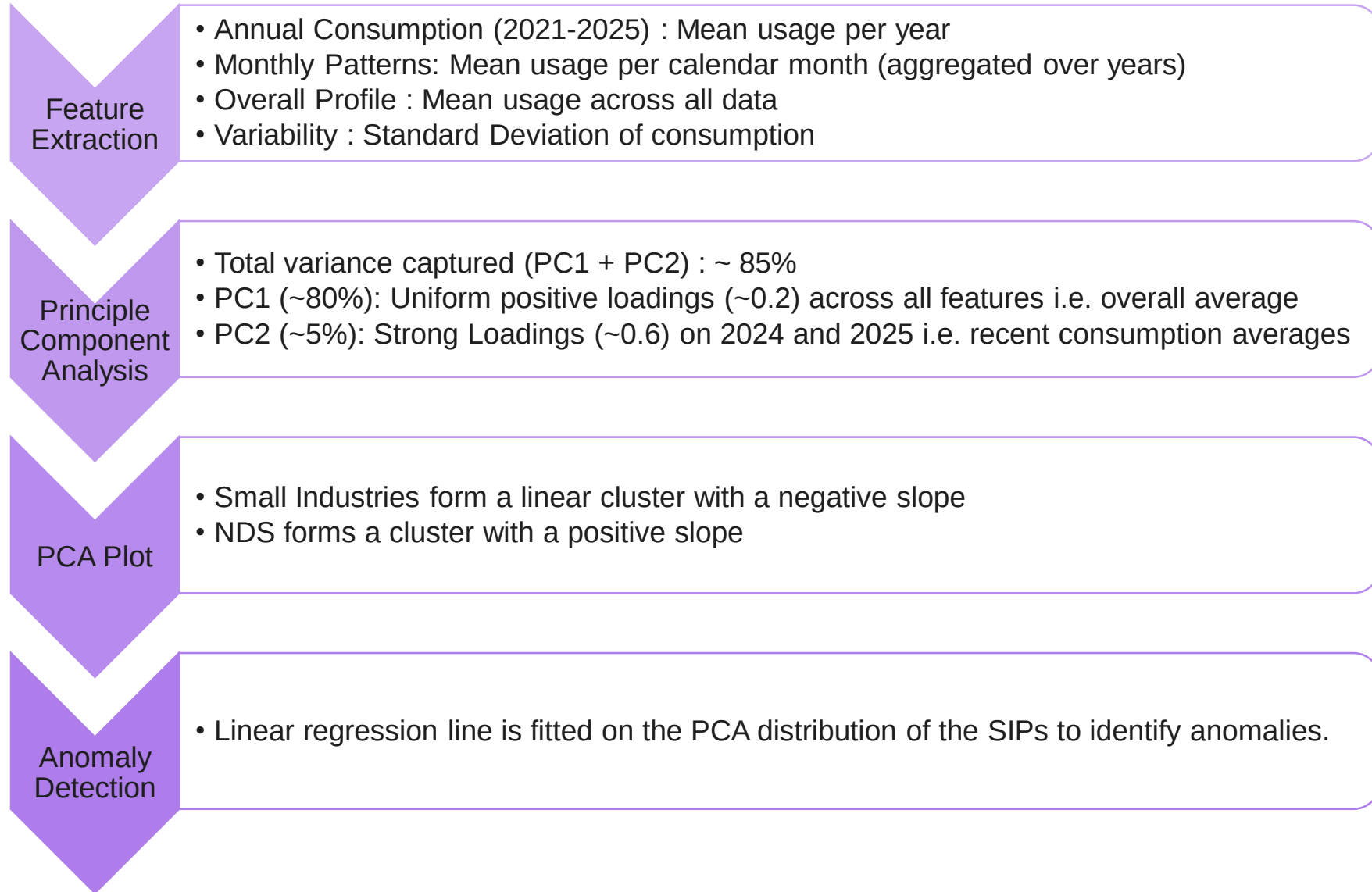
## STEP 04



# Monthly Consumption Trends

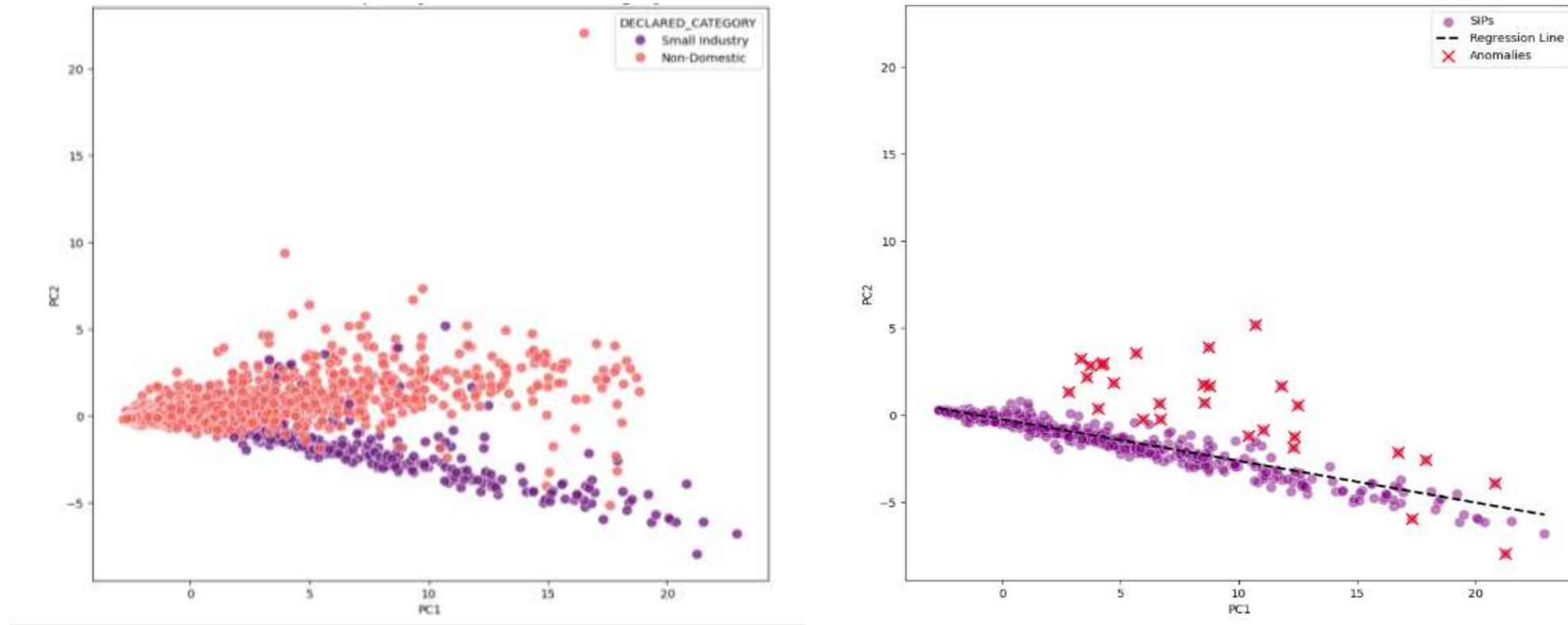


# II. Consumption Based Detection



$$5 + 12 + 1 + 1 = 19 \text{ features}$$

# PCA Plot



The consumption of Non Domestic establishments have increased in the recent years while that of Small Industries have shown a decreasing trend.

# III. Address Based Detection



# Output Snippet

|   | Name<br>GMAP                                     | Phone<br>Number<br>GMAP | Search<br>Radius | Nearest NDS<br>Name               | Nearest<br>Reg. NDS<br>Mobile No. | SIP Name                               | Reg. SIP<br>Mobile No. | Distance<br>to SIP | NDS<br>Match<br>Flag | SIP<br>Match<br>Flag |
|---|--------------------------------------------------|-------------------------|------------------|-----------------------------------|-----------------------------------|----------------------------------------|------------------------|--------------------|----------------------|----------------------|
|   | Hindustan<br>Petroleum<br>Corporation<br>Limited | 9460132963              | 5.24             | HPCL SHIVAM<br>FILLING<br>STATION | 9460132963                        | m/s balaji<br>marble and<br>kota stone | 9252059456             | 3.49               | True                 | False                |
|   | Hotel Ridhi<br>Sidhi Palace                      | 9252901702              | 29.44            | GANESH<br>GOYAL                   | 9252901702                        | Dhan Kumar<br>Singodiya                | 9828111224             | 24.01              | True                 | False                |
| ! | Bharat<br>Library                                | 8209270735              | 59.01            | ATCTELECOMI<br>NFRA<br>STRUTURE   | 9929106687                        | SALASAR<br>EMBROIDERY                  | 8209270735             | 16.95              | False                | True                 |
|   | Hotel Shyam<br>Sharnam                           | 9928138169              | 15.16            | SMT. KRISHNA<br>JAISWAL           | 9928138169                        | sahu floor mil                         | 9269141516             | 5.67               | True                 | False                |
| ! | Delhi Cafe                                       | 8107306700              | 25.71            | SADIQ MIYA                        | 9251786216                        | Smt,Shanti<br>Devi                     | 8107306700             | 16.81              | False                | True                 |



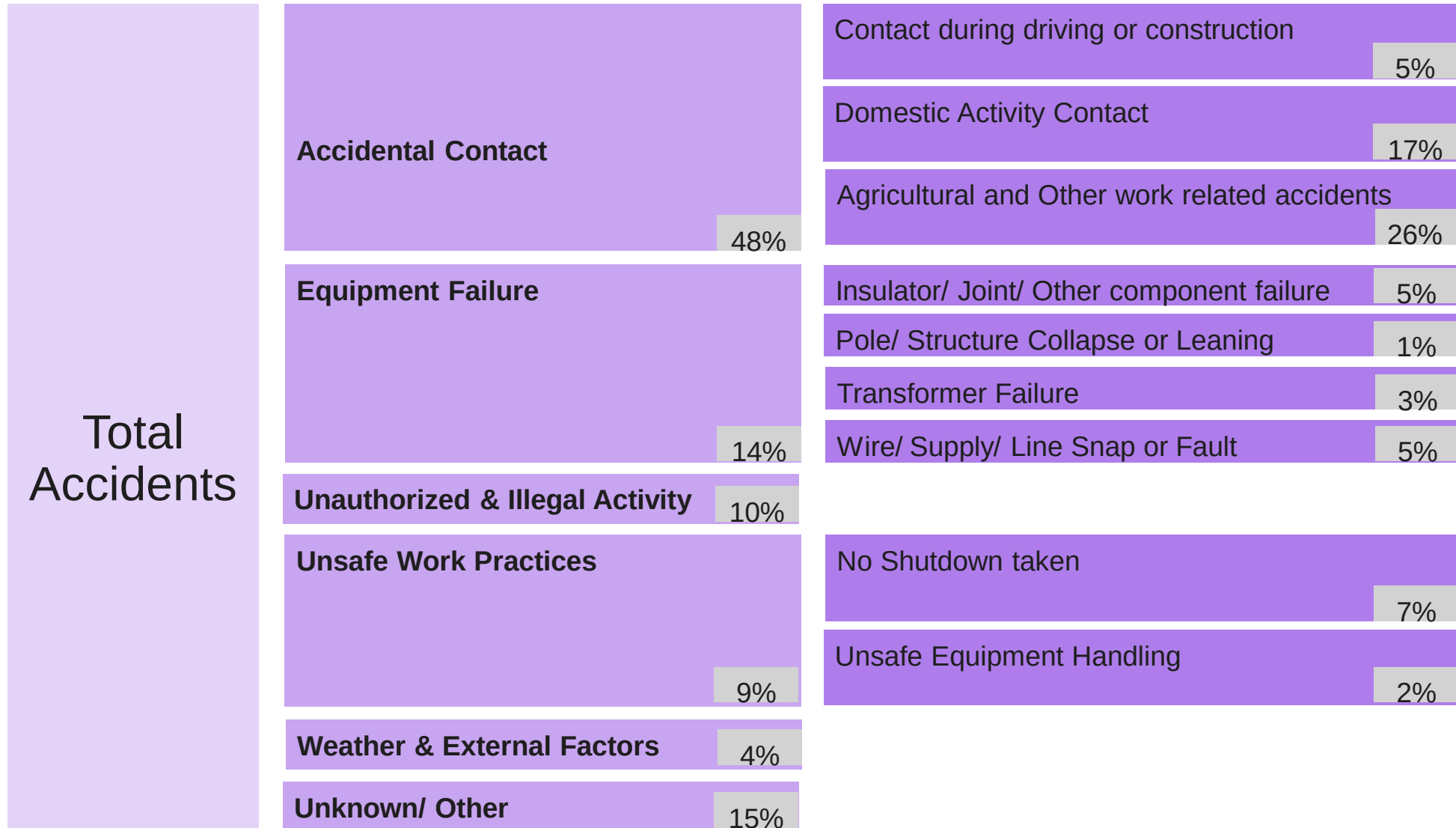
# Future Prospects

|                                                   |                                                                                     |
|---------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>Alternative Ownership Identification</b>       | Explore methods beyond Google Maps to reliably retrieve owner information           |
| <b>Process Documentation &amp; GitHub updates</b> | Maintain well documented and updated codebase for transparency and collaboration    |
| <b>Scalability Across Districts</b>               | Extend the solution to other districts (Banskho, Alwar)                             |
| <b>Automation &amp; Continuous Monitoring</b>     | Reduce manual intervention with continuous, automated detection                     |
| <b>Real Time Theft Detection Dashboard</b>        | Provide clear, real-time visibility of suspected thefts via a centralized dashboard |

# Project 2 Overview: Cause Analysis

| Topics                    | Sub-Topics Covered                                                                                                                                             |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Preparation          | <ul style="list-style-type: none"><li>• Converted Krutidev and Devanagari scripts to their English translations for further processing.</li></ul>              |
| Supervised Classification | <ul style="list-style-type: none"><li>• DeepSeek API with batch processing</li><li>• Prompt-based classification</li><li>• Six main cause categories</li></ul> |
| Sub-Categorization        | <ul style="list-style-type: none"><li>• Divided into further sub-classes</li></ul>                                                                             |
| Accident Dashboard        | <ul style="list-style-type: none"><li>• Streamlit web app enabling filtering, trend analysis and visualization of accident data.</li></ul>                     |

# Accident Classification



# Accident Dashboard

